Yiyang Huang

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Summary

Ph.D. student at Northeastern University focusing on long-video understanding, hallucination detection and mitigation, and reflective, tool-using, self-improving agents, with research internships at Adobe Research and first-author publications at ICLR, ECCV, ACL, and EMNLP.

Education

Northeastern University — Boston, MA
Ph.D. in Computer Engineering, advised by Prof. Yun Raymond Fu — *Sep 2024 – Aug 2029 (Expected)*

Xidian University — Xi'an, China
M.S. in Computer Science, advised by Prof. Xuefeng Liang — *Sep 2021 – Jun 2024*
B.Eng. in Intelligence Science and Technology — *Sep 2017 – Jun 2021*

Research Experience

Northeastern University — Boston, MA
Ph.D. student — Sep 2024 – Present

[*Preprint*] Reflective Frame Selection Agent for Long-Video Understanding
Focus Areas: Long-Video Understanding; Task Decomposition; Reflective Agents; Tool-Augmented Retrieval
- Developed RACER, a training-free reflective agentic framework that decomposes long-video frame selection into query interpretation by a lightweight Vid-LLM and evidence localization through an embedding-based retrieval tool.
- Improved performance by 4.3–6.6 points over uniform sampling across Video-MME, MLVU, and LongVideoBench with four Vid-LLMs, with gains up to 11.6 points on MLVU under a 32-frame answering budget.

[ICLR'26] Mitigating Visual Hallucinations in MLLMs
Focus Areas: Hallucination Mitigation; Visual Encoder Bias; Inference-Time Intervention
- Identified encoder-side causes of hallucinations and proposed SHIELD, an inference-time intervention combining token reweighting/subtraction with adversarial contrastive decoding, achieving a 27% relative reduction in captioning hallucinations across multiple MLLM families.

[EMNLP'25] Training-Free Adaptation of Image-Pretrained MLLMs for Video Understanding
Focus Areas: Efficient Video Understanding; Token Compression; Question Decomposition
- Developed D-CoDe, a plug-and-play framework that adapts image-pretrained MLLMs for VideoQA through question decomposition and dynamic token/frame compression, achieving a 29% relative improvement in egocentric VideoQA accuracy with LLaVA-NeXT-7B.

Industry Experience

Adobe Research — San Jose, CA
Research Intern — May 2026 – Nov 2026 (Expected) | May 2025 – Nov 2025

[Ongoing] Multimodal Agents for Visual Evaluation and Self-Improvement
Mentors: Dr. Zhaowen Wang, Dr. Shraman Pramanick
Focus Areas: Multimodal Agents; Multi-Agent Collaboration; Self-Improvement
- Constructed design-evaluation data and post-trained Qwen3.5-9B with GRPO and LoRA to learn implicit aesthetic criteria through independent scoring and structured visual reasoning.
- Designed a self-improvement loop pairing a memory-augmented VLM evaluator with a Codex editing agent, using before/after score gains to define rewards for multi-turn GRPO.
- Achieved 84.3% mean pairwise accuracy on Crello and internal Adobe product data, outperforming Gemini-3.6-Flash (73.0%), Claude-Opus-4.8 (66.9%), and GPT-5.6-Sol (59.0%).

[ECCV'26] [Patent Filed] Compositional Layout Understanding in MLLMs
Mentors: Dr. Zhaowen Wang, Dr. Simon Jenni, Dr. Jing Shi
Focus Areas: Vision-Language Models; Compositional Reasoning; Structured Visual Understanding
- Identified semantic drift and structural ambiguity as key challenges in compositional visual reasoning, and developed MASON with metadata-visual grounding and 3D structural perception, achieving 26% and 15% relative improvements over Gemini-2.5-Pro and GPT-5/GPT-o3, respectively, using Qwen2.5-VL-7B as the backbone.
- Constructed CoDeLayout, a ~20K-sample benchmark, via an automated pipeline for compositional structure mining and multimodal QA generation.

Publications

First-Author Papers

- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Jianglin Lu, Qihua Dong, Hailing Wang, Huimin Zeng, Mingyuan Zhang, Yun Fu. "RACER: Reflective Agent Coupling Query Interpretation and Tool-Based Retrieval for Frame Selection in Long Video Understanding." ***Preprint***. [Paper]
- **Yiyang Huang**, Zhaowen Wang, Simon Jenni, Jing Shi, Yitian Zhang, Yizhou Wang, Yun Fu. "Beyond Atomic Layouts: Compositional Design Understanding with Vision-Language Models." **ECCV 2026**. [Paper] [Code] [Project Page]
- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Mingyuan Zhang, Liang Shi, Huimin Zeng, Yun Fu. "Distorted or Fabricated? A Survey on Hallucination in Video LLMs." **ACL 2026** (Findings). [Paper] [Repo]
- **Yiyang Huang**, Liang Shi, Yitian Zhang, Yi Xu, Yun Fu. "SHIELD: Suppressing Hallucinations in LVLM Encoders via Bias and Vulnerability Defense." **ICLR 2026**. [Paper] [Code]
- **Yiyang Huang**, Yizhou Wang, Yun Fu. "D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression and Question Decomposition." **EMNLP 2025** (Main). [Paper] [Code]
- **Yiyang Huang**, Xuefeng Liang, Chaowei Fang. "CALLip: Lipreading using Contrastive and Attribute Learning." **ACM MM 2021**. [Paper]

Selected Co-Authored Papers

- Hailing Wang, Jianglin Lu, Xu Ma, **Yiyang Huang**, Yun Fu. "StreamVLM: Adapting Offline Vision-Language Models for Streaming Understanding." ***Under review***.
- Mingyuan Zhang, Yue Bai, Yifan Wang, **Yiyang Huang**, Yun Fu. "Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models." ***Preprint***. [Paper]
- Shuai Zou, Xuefeng Liang, **Yiyang Huang**. "LipReading for Low-resource Languages by Language Dynamic LoRA." **ICASSP 2025**. [Paper]

Academic Service

Conference Reviewing: ICLR; NeurIPS; ACL; EMNLP; WWW; IEEE Face & Gesture (FG)
Journal Reviewing: ACM TKDD

Teaching

Teaching Assistant

DS 5110: Essentials of Data Science — Fall 2025
DS 5020: Fundamentals of Linear Algebra and Probability — Spring & Fall 2026

Skills

Languages: Python, LaTeX
ML Frameworks: PyTorch, Hugging Face Transformers, Hugging Face Datasets
Training/Fine-Tuning Libraries: Accelerate, PEFT, bitsandbytes
RL/Post-Training Frameworks: TRL, verl, OpenRLHF
Inference & Evaluation: vLLM, lmms-eval
Video Processing: ffmpeg, PyAV, decord
Systems/Infrastructure: Linux, Git, Slurm

Awards

National Scholarship, China — 2021
Outstanding Student, Xidian University — 2022
1st Prize, Undergraduate Computer Design Competition (National Level), China — 2021
2nd Prize, RoboMaster National Robotics Competition, China — 2019
3rd Prize, ICRA AI Challenge — 2019